1 Correspondence to Cluster Size Distribution Formula

The mathematical formula for computing the average cluster size distribution across N mice is:

$$y(x) = \frac{1}{N} \sum_{i=1}^{N} n_i(x),$$

where $n_i(x)$ denotes the number of events of size x (e.g., number of pellets) for mouse i.
In your Python code, this logic is expressed in the function `average_cluster_sizes_across_mice`, shown below:

Listing 1: Computing the average cluster size distribution

```python
def average_cluster_sizes_across_mice(pellettimes_group, max_pellets_per_event=10):
    """
    Calculate the average cluster size distribution across multiple mice.

    Parameters
    ----------
    pellettimes_group : list of lists
        A list of timestamp lists for each mouse.
    max_pellets_per_event : int
        The maximum number of pellets per event to include in the analysis.

    Returns
    -------
    avg_cluster_sizes : list
        The average number of events for each cluster size (1 pellet, 2 pellets, etc.).
    """
    # Step 1: Initialize counts for each pellet size.
    total_cluster_counts = np.zeros(max_pellets_per_event)

    # Step 2: Loop over each mouse and accumulate cluster sizes.
    for pellettimes in pellettimes_group:
        cluster_sizes = get_cluster_size_distribution_for_mouse(pellettimes)
        for size in cluster_sizes:
            if size <= max_pellets_per_event:
                total_cluster_counts[size - 1] += 1
            else:
                # If size exceeds max_pellets_per_event, add to the last bin
                total_cluster_counts[-1] += 1

    # Step 3: Divide by the number of mice to get the average across mice.
    avg_cluster_sizes = total_cluster_counts / len(pellettimes_group)

    return avg_cluster_sizes
```

Explanation:

- Within the `for` loop, $n_i(x)$ is incremented each time a cluster of size x is found for mouse i.
- `total_cluster_counts`$[x-1]$ thus accumulates $\sum_{i=1}^{N} n_i(x)$, the total counts of size x clusters across all N mice.

- Dividing by `len`(*pellettimes_group*) is equivalent to dividing by N, producing $\dfrac{1}{N}\sum_{i=1}^{N} n_i(x)$, which is $y(x)$, the **average** number of events of size x across mice.